Dataset Overview (Metadata)

We will use the Wine Quality Dataset (WineQT.csv), a well-known dataset from the UCI Machine Learning Repository, also available on Kaggle.

The dataset contains physicochemical properties of red wine samples along with a quality score assigned by wine tasters.

Features (all numeric):

1. fixed acidity – Tartaric acid (g/dm^3)
2. volatile acidity – Acetic acid (g/dm^3)
3. citric acid – Citric acid (g/dm^3)
4. residual sugar – Remaining sugars after fermentation (g/dm^3)
5. chlorides – Sodium chloride (g/dm^3)
6. free sulfur dioxide – Free SO_2 (mg/dm^3)
7. total sulfur dioxide – Total SO_2 (mg/dm^3)
8. density – Density of the wine (g/cm^3)
9. pH – Acidity level (0–14 scale)
10. sulphates – Potassium sulphate (g/dm^3)

Target Variable:

quality – Integer rating of wine quality (range 3–8).

Objective of the Exercise

The goal is to predict wine quality based on its chemical properties using K-Nearest Neighbors (KNN) algorithm.

Since the target variable quality is an integer (ordinal scale), we can approach it in two ways:

1. Classification Task: Treat wine quality as categorical (low/medium/high).
2. Regression Task: Predict the numeric quality score directly.

You will practice both approaches.

Exercise Instructions

Step 1: Load and Explore the Data

- Load the WineQT.csv dataset into a DataFrame.
- Display the first 5 rows.
- Check shape (rows, columns).
- Compute basic statistics (.describe()).
- Plot histograms of features to understand their distributions.

Step 2: Preprocessing

- Check for missing values.
- Since all features are numeric, no encoding is required.
- Perform feature scaling (StandardScaler or MinMaxScaler) — important for KNN since it is distance-based.

Step 3: Train-Test Split

- Split dataset into training (80%) and testing (20%).

Step 5: KNN Classification

1. Train a KNN Classifier with different values of k (neighbors).

2. Evaluate using accuracy, confusion matrix, and classification report (precision, recall, F1).
3. Plot accuracy vs k to choose the optimal k. (Use matplotlib library)

👉 Explain: Why does accuracy change with different k values?

Step 6: KNN Regression

1. Train a KNN Regressor with different values of k.
2. Evaluate using R^2, RMSE, and MAE.
3. Plot error metrics vs k. (Use matplotlib library)

👉 Explain: Why do smaller values of k tend to overfit while larger values smooth predictions?

Step 7: Compare Results

- Which approach (classification or regression) seems more meaningful for this dataset?
- Do wines with higher pH/sulphates tend to get higher quality ratings?

Learning Outcomes

By completing this exercise, you will:

- Understand how KNN works for both classification and regression.
- Learn the importance of scaling in distance-based algorithms.
- Gain experience in model evaluation and hyperparameter tuning.